

HealthConnect Clinic Experience Lab

Week 5 - Exploratory Data Analysis and Key Findings

Introduction

My Week 5 HealthConnect Clinic Experience Lab moved me from the planning stage in Week 4, into the actual analysis of the appointment dataset. The main focus of my analysis is to better understand appointment attendance and no-show patterns, so as to identify factors that may be associated with patients missing their appointments.

I worked with the HealthConnect Appointment Dataset containing 5,000 appointment records and used exploratory data analysis to examine appointment outcomes in relation to reminders, appointment type, previous no-shows, appointment time and day, age group, gender, booking lead time, distance to the clinic and waiting time.

Week 4 to Week 5 Transition

In Week 4, I focused on understanding the HealthConnect appointment problem and how the available data could help me examine appointment attendance and no-show patterns. I reviewed the Appointment Dataset and Data Dictionary, identified the business questions and proposed KPIs that would guide my analysis. I also noted the missing values in distance to clinic and waiting time, and that any relationship found in the data should not automatically be treated as the cause of a no-show. My plan for Week 5 was therefore to prepare the data, carry out deeper analysis and begin developing the dashboard.

➤ Data Preparation

Before beginning the analysis, I created a separate working copy of the appointment dataset so that the original dataset will remain unchanged. I then reviewed the variables, missing values, duplicate records and data consistency. I also checked the data types and duplicate appointment IDs. I found no duplicate appointment IDs, while the date fields required closer attention because of the inconsistencies I identified.

Just as the initial assessment, there were 90 missing values in the distance-to-clinic field and 60 missing values in the waiting-time field. I retained the missing values, rather than just replacing them with assumptions.

I also observed inconsistent date representations in the booking date and appointment date fields. As I attempted to standardise the dates, it revealed inconsistencies when compared with the supplied booking lead days. I retained the original date values and used the provided `booking_lead_days` variable for lead-time analysis rather than extracting them from the date fields.

I then decided to use the cleaned working dataset for the exploratory analysis, while preserving the original dataset.

➤ Exploratory Data Analysis

I used PivotTables in Excel to explore the appointment data and compare appointment outcomes across different patient and appointment factors. My main interest was to understand the patterns around no-shows and identify factors that may be associated with patients missing their appointments.

I analysed overall appointment outcomes and compared them with reminder status, reminder channel, appointment type, previous no-show history, appointment time, appointment day, age group and gender. I also compared the average booking lead time, distance to the clinic and waiting time across the different appointment outcomes.

➤ Key Findings

i. Overall Appointment Outcomes

Out of the 5,000 appointments analysed, 2,423 were no-shows, 2,314 were attended and 263 were cancelled. This represents a no-show rate of approximately 48.5%, compared with an attendance rate of 46.3% and a cancellation rate of 5.3%, which shows that almost half of the appointments in the dataset resulted in patients not showing up.

ii. Reminder Status

Among the appointments where a reminder was sent, 1,721 out of 3,634 resulted in no-shows - this represents approximately 47.4%. For appointments where no reminder was sent, 702 out of 1,366 resulted in no-shows, representing approximately 51.4%. This shows a lower no-show rate among appointments where reminders were sent.

iii. Previous No-Show History

Previous no-show history showed one of the clearest patterns in the analysis. Appointments involving patients with no previous no-shows had a no-show rate of approximately 43.5%. This increased to 53.5% for those with one previous no-show, 59.4% for those with two, and 67.9% for those with three previous no-shows. What this suggests is that a patient's previous no-show history may be useful in identifying appointments with a higher risk of being missed.

iv. Booking Lead Time

Booking lead time showed a noticeable difference across appointment outcomes. Patients who attended their appointments had an average booking lead time of 24.52 days, while cancelled appointments averaged 29.63 days. No-show appointments had the highest average booking lead time at 34.53 days. This means that no-show appointments were booked about 10 days further in advance than attended appointments on average. By this, booking lead times may be associated with a higher likelihood of patients missing their appointments.

v. Appointment Type

Follow-up appointments recorded the highest no-show rate at approximately 51.2%, followed by diagnostic tests at 49.7%. Specialist consultations had a no-show rate of 47.4%, while general

consultations recorded the lowest rate at approximately 46.6%. And although the differences are not extremely large, follow-up appointments appear to have a slightly higher no-show risk.

vi. **Reminder Channel**

The analysis also showed differences across reminder channels. Appointments with SMS reminders had the lowest no-show rate at approximately 45.8%, compared with 48.4% for email and 49.8% for WhatsApp. Appointments with no reminder channel had the highest no-show rate at approximately 51.4%. What this means is that reminder communication, particularly SMS in this dataset, may be associated with better appointment attendance. However, the analysis only shows an association and does not prove that the reminder channel itself caused the difference.

vii. **Distance to the Clinic**

Patients who attended, travelled an average distance of 9.67 km to the clinic, compared with 10.53 km for no-show appointments. The difference is relatively small, which suggests that distance may have some relationship with appointment attendance, but it does not appear to be one of the strongest patterns identified in this analysis.

viii. **Waiting Time**

Average waiting time was almost the same for attended and no-show appointments. Attended appointments recorded an average waiting time of 24.29 minutes, while no-shows averaged 24.20 minutes. Going by this comparison, waiting time does not show a meaningful difference between the two groups.

➤ **KPI Development**

i. **No-Show Rate**

This KPI measures the percentage of scheduled appointments recorded as No-Show. It is important because it helps to show the extent of HealthConnect's missed-appointment problem and it answers Business Question 1. The no-show rate is 48.46%, which means that almost half of the 5,000 appointments resulted in no-shows.

ii. Reminder No-Show Rate

This KPI compares the no-show rate for appointments where a reminder was sent with those where no reminder was sent. It helps me understand whether reminder use may be associated with appointment attendance and it answers Business Question 2. The no-show rate was 47.36% where reminders were sent, compared with 51.39% where no reminder was sent. This suggests that reminders may be associated with a lower no-show rate.

iii. Average Booking Lead Time for No-Shows

This KPI measures the average number of days between booking and appointment date for appointments recorded as No-Show. It answers Business Question 3 and helps to understand whether longer booking periods may be associated with missed appointments. The average booking lead time for no-shows was 34.53 days, compared with 24.52 days for attended appointments. This suggests that longer booking lead times may be associated with patients missing their appointments.

iv. Repeat No-Show Rate

This KPI measures the no-show rate among patients who already had at least one previous no-show. It answers Business Question 4 and helps to understand whether previous attendance behaviour may be useful in identifying patients at a higher risk of missing another appointment. The repeat no-show rate was 55.41%, which is higher than the overall no-show rate of 48.46%.

➤ Initial Interpretation

This exploratory analysis shows that HealthConnect's no-show problem is not linked to just one factor. However, some patterns stand out more clearly than others. Previous no-show history and longer booking lead times appear to be particularly important factors associated with missed appointments.

Reminders also show a useful pattern, as appointments where reminders were sent had a lower no-show rate than those without reminders. Among the reminder channels, SMS recorded the lowest no-show rate. Appointment type also showed some variation, with follow-up appointments recording the highest no-show rate.

On the other hand, distance to the clinic showed only a small difference between attended and no-show appointments, while waiting time showed almost no difference. These findings help narrow the areas that may deserve more attention when considering ways to reduce missed appointments.

➤ **Business Recommendations**

Based on the findings from my analysis, HealthConnect Clinic should continue using appointment reminders because appointments where reminders were sent had a lower no-show rate than those without reminders.

More attention could also be given to patients with a history of previous no-shows since this group recorded a higher no-show rate. Appointments booked far in advance may also require additional follow-up reminders because no-show appointments had a longer average booking lead time.

The clinic may also pay closer attention to follow-up appointments because they recorded the highest no-show rate among the appointment types analysed.

➤ **Limitations**

The findings in this analysis show patterns and relationships in the dataset, but they do not prove that any of the factors directly caused patients to miss their appointments.

There were also some missing values in the distance-to-clinic and waiting-time fields. These values were retained rather than replaced with assumed values.

The analysis is also limited to the variables available in the HealthConnect dataset, so there may be other factors affecting appointment attendance that are not included in the data.

➤ **Assumptions, Risks and Dependencies**

The analysis depends on the accuracy and completeness of the HealthConnect dataset provided. The missing values identified in distance to clinic and waiting time remained a limitation during Week 5, while the inconsistency in the date fields meant that I relied on the provided `booking_lead_days` variable for my lead-time analysis.

Another important consideration is that the patterns I identified in the data show relationships but do not confirm that any of the factors directly caused patients to miss their appointments. Some grouped values, particularly for distance and waiting time, may also contain fewer appointments than others, so extreme rates in some groups should be interpreted carefully.

No major assumptions or dependencies from Week 4 were resolved or changed during Week 5, while the date inconsistency was an issue that became clearer during the analysis and was addressed by using the supplied `booking_lead_days` variable.

➤ Conclusion

In conclusion, Week 5 analysis has helped me move beyond simply understanding the dataset to identifying patterns within it. With almost half of the appointments resulting in no-shows, the findings show that missed appointments are an important issue for HealthConnect Clinic.

And based on the exploratory analysis, previous no-show history, booking lead time, reminders and appointment type are some of the main areas that need to be investigated further. These findings are descriptive and should not be interpreted as proof that any one factor causes patients to miss appointments.

The next stage of the analysis will focus on refining the findings further and building on the evidence from the data as the HealthConnect project moves into the next phase.